Can Generative LLMs Help Classify Imbalanced Real-World Data ? Exploring Rare Diseases on Social Media

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KCNB1 encephalopathy: a very rare and severe disease

Developmental and Epileptic Encephalopathies (DEE): group of rare and severe diseases which commonly begin in infancy or childhood. They are associated with frequent seizures of different types, intellectual disability and developmental delay, plateau or regression.

KCNB1 encephalopathy :

- Only known since 2014
- Wide phenotypic sprectrum
- Diverse symptoms including drug-resistant epileptic crises, spasms, severe intellectual disability, autistic disorders, developmental delays, anxiety, sleep and eating disorders, hyperactivity, …

In France, in 2023:

- 18% of the population
- 26% of the caregivers
- 30% of parents of children younger than 15 years old

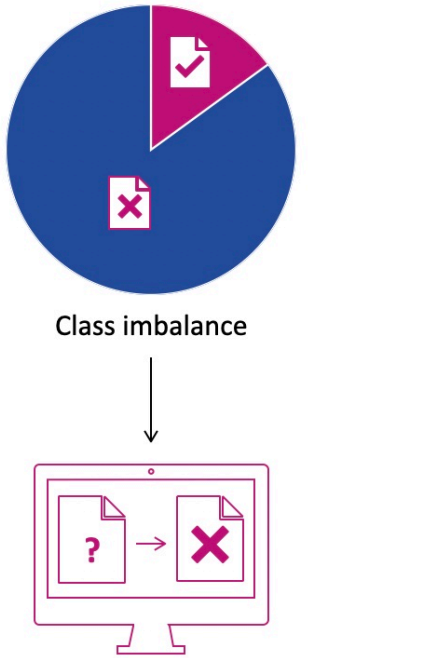
} *post at least occasionally online about their health or someone close's*

Objective

Correctly detect messages that relate patients' or their caregivers' experience of a DEE within a dataset of social media messages containing at least one term associated with a DEE and written in French.

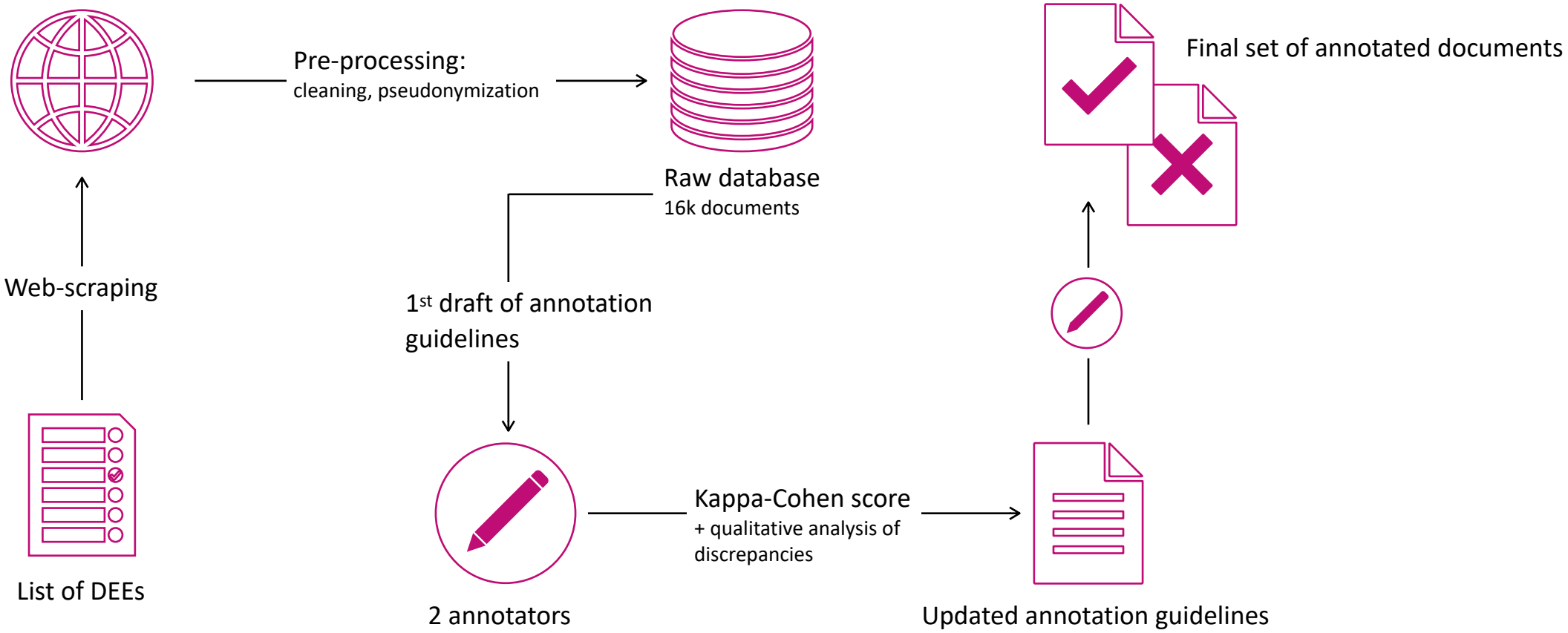
Journée mondiale du syndrome de Dravet | ⚡ Plus de 50 enfants atteints par cette épilepsie sévère sont suivis par le centre des épilepsies rares à tag_nom tag_nom 🙏 Merci aux équipes très impliquées dans leur prise en charge et dans la recherche autour de cette maladie !

tag_nom Courage, je comprends parfaitement, papa d'une petite fille de 2 ans et demi avec suspicion de syndrome de Rett c'est très dur. Elle ne marche pas, ne pointe pas du doigt, n'a jamais dit papa. Il faut arriver à oublier toutes les projections qu'on s'est fait durant la grossesse.



- *Could (a small and open-source) generative LLMs help tackle the imbalance issue of this classification problem? And/or reduce human annotation needs?*
- *By generating minority class samples ? Or by classifying directly as a 0-shot task ?*

Data & annotation



Experimental setup

Baseline model

- CamemBERT fine-tuned on our classification task
- 3 epochs / batch size: 8 / weight decay: 0.01

Evaluation

- 10 runs with different random seeds to split data between test and training/validation
- Precision, recall, and F1-score on the positive class and on macro-average
- Mean and 95% CIs of these metrics on the 10 runs

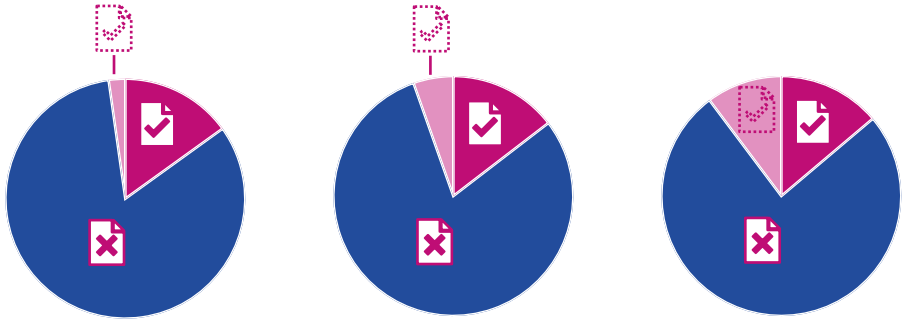
→ *use of Llama3.2 to reduce class imbalance or annotation needs ?*

Prompting strategies

- *prompt_en1*: English prompt, mentioning DEE
- *prompt_en2*: English prompt, mentioning only "diseases"
- *prompt_fr*: French prompt, mentioning only "diseases"

Two oversampling strategies (2%, 5%, 10% of training data)

- *completion*: messages from the minority class were truncated and the prompt contained an instruction to complete the message with two sentences
- *new*: the prompt contained an instruction to write another message, similar to the one given as an example



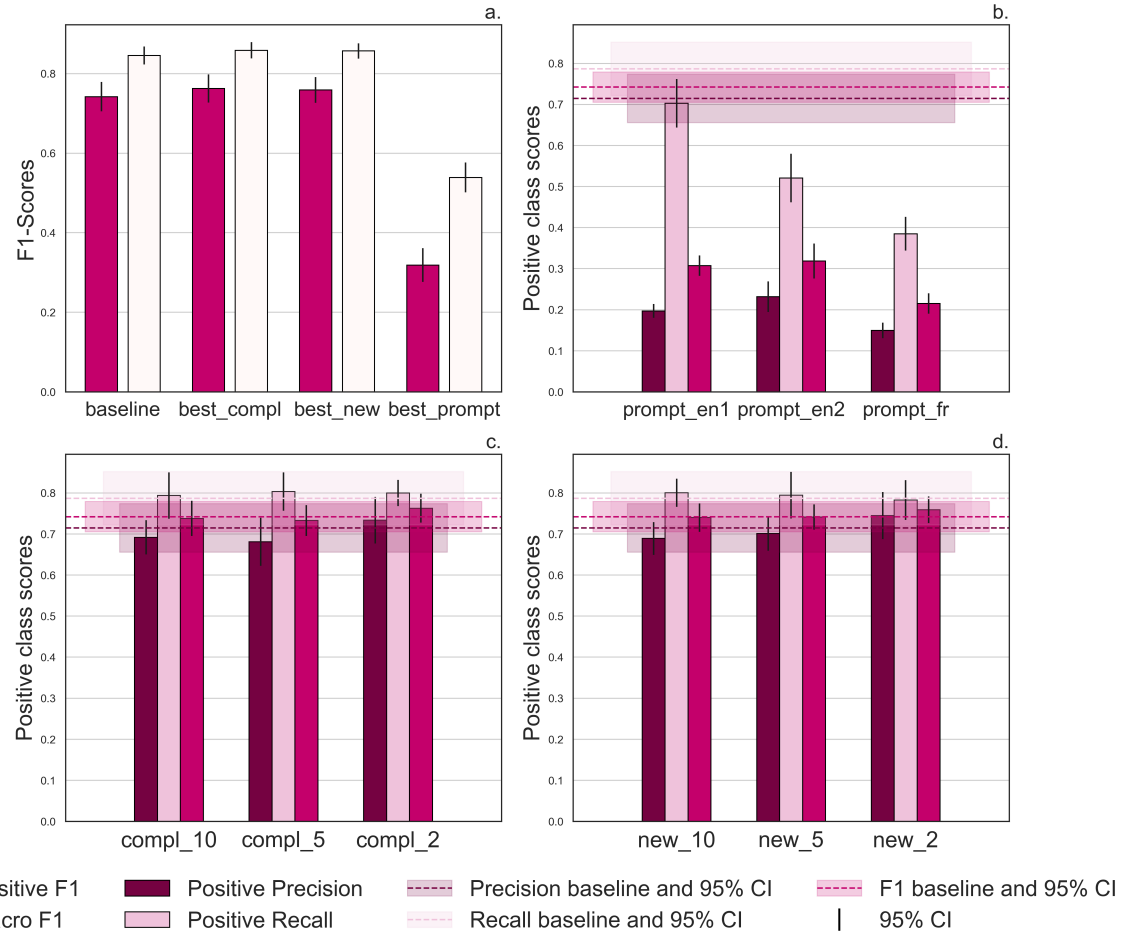
Martin L, Muller B, Suárez PJO, Dupont Y, Romary L, de la Clergerie ÉV, et al. CamemBERT: a Tasty French Language Model. Proc. 58th Annu. Meet. Assoc. Comput. Linguist., 2020, p. 7203–19. https://doi.org/10.18653/v1/2020.acl-main.645.
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Results

Annotation process: 1,095 annotated document with a final Kappa-Cohen score of 78%

Baseline
- Macro F1: 84.57%
- Positive class F1: 74.21%
- Positive precision: 71.47%
- Positive recall: 78.68%

Results of the experiments
- Significant drop of positive/macro F1 (31.87% / 53.89%) when using 0-shot prompting
- Only a very slight improvement of positive/macro F1 (76.28% / 85.87%) with data augmentation strategies
- Best prompting strategy: the one without mention of DEE and in English
- Data augmentation strategies: setups with 2% generated data achieve the best results, improving positive precision (74.48%) compared to baseline. With 10%, positive precision drop.



Conclusions

Key take aways

- Improvement of balance between positive recall and precision with low volume of generated data
- But smaller improvement than in the literature
- Very rare diseases + social media data → increased complexity of the tasks

Limitations

- Small-sized LLM
- Ethics and regulatory aspects
- Inter-annotator agreement could be better
- Societal and environmental impacts of LLMs not to be forgotten

Next steps and perspectives

- Short term:
  - Model was applied on data from KCNB1 Facebook group : 1/3 are experiences of the disease
  - Clustering and topic exploration on these posts
  - Multi-view contrastive learning to improve representation of this particular data to have a better clustering
- Long term objective: better understanding this very rare disease by incorporating social media data and EHR data together

Münker S, Kugler K, Rettinger A. Zero-shot prompt-based classification: topic labeling in times of foundation models in German Tweets 2024. https://doi.org/10.48550/arXiv.2406.18239.
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Clinical Bioinformatics & Data Science teams

Full paper

Thank you !

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